activation along a one-dimensional line in representation space, $\tilde{h}_l(m) = h_l + m\,\Delta h$. Under the manifold hypothesis, degradation is governed primarily by how far this line trajectory departs from the typical region near $\mathcal{M}_l$, so it is natural to model $D(m)$ as a smooth function of the (signed) distance along this line to the nearest "on-manifold" locations. In particular, as illustrated in Fig. 3, the steered trajectory may intersect the manifold neighborhood at one or more values $\{m_i\}$ (e.g., one for $m > 0$ and one for $m < 0$). We therefore model validity as being highest near these intersection points and decaying as $|m - m_i|$ grows.

A convenient choice that is positive, smooth, and exhibits heavy-tailed distance-based decay is the rational quadratic (RQ) form, widely used in kernel methods and Gaussian processes to model multi-scale, polynomial-rate attenuation with distance (Rasmussen, 2004). Prior research on controllability metrics has established that model steerability is often asymmetric (Miehling et al., 2025), exhibiting varying degrees of responsiveness along different directions of the same dimension. Motivated by this observation, we employ a piecewise parameterized model to quantify degradation:

$$D(m) = \begin{cases} \left(1 + \frac{(m-m_+)^2}{L_+}\right)^{-p_+} & \text{if } m \geq 0 \\ \left(1 + \frac{(m-m_-)^2}{L_-}\right)^{-p_-} & \text{if } m < 0 \end{cases} \tag{12}$$

where $m_\pm$ corresponds to the signed distance from the original activation point $P$ to an on-manifold intersection point $P_\pm$ along the steering line (Fig. 3); $L_\pm$ sets the characteristic scale of decay and reflects how fast the distance-to-manifold grows along the steering direction (larger when the direction is locally parallel to the manifold and smaller when it cuts across it); and $p_\pm$ controls the decay rate (tail heaviness) as the trajectory moves away from the manifold neighborhood.

### 4.2 Preference Capability: Projection Gain With Decay

We study how additive steering changes a model's preference through intermediate activations. An intervention at layer $l$ updates the hidden state as $\tilde{h}(m) = h + m\,\Delta h$.

Prior work under LRH-style assumptions often models preference probability with a logistic form, $P(p_p \mid h) = \sigma\big(-(\boldsymbol{\omega}_p^\mathsf{T} h + b_p)\big)$, where $\boldsymbol{\omega}_p$ is the preference vector. Separately, work on activation geometry suggests that after low-dimensional projection (e.g., PCA), opposite preference labels are often approximately linearly separable. Under the activation-manifold view, this motivates a two-dimensional *preference plane* and a preference direction whose signed coordinate reflects *preference intensity*. Our contribution is to incorporate validity attenuation $D(\cdot)$ (Assumption 4.2) to account for off-manifold steering.

To model this, we write the steered preference probability as

$$P(p_p \mid \tilde{h}(m)) = \sigma\Big(-\big(\boldsymbol{\omega}_p^\mathsf{T} h + \alpha_p m\big) D_p(m) - b_p\Big), \tag{13}$$

where $\alpha_p \triangleq \boldsymbol{\omega}_p^\mathsf{T} \Delta h$ measures how much the steering direction aligns with the preference vector: $\alpha$ is large when $\Delta h$ is aligned with $\boldsymbol{\omega}_p$, and $\alpha_p = 0$ when $\Delta h$ is orthogonal to $\boldsymbol{\omega}_p$.

This implies the preference log-odds

$$\log \frac{P(p_p \mid \tilde{h}(m))}{1 - P(p_p \mid \tilde{h}(m))} = \big(\boldsymbol{\omega}_p^\mathsf{T} h + \alpha_p m\big) D_p(m) + b_p. \tag{14}$$

**Key implication (linear regime → nonlinear collapse).** From Eq. (14), the $m$-dependence enters as $\alpha_p m\, D_p(m)$. When $|m - m_\pm| \ll L_\pm$, Eq. (12) gives $D_p(m) \approx 1$, hence preference log-odds is approximately linear in $m$ with slope $\alpha$ (matching the near-linear regime in Bigelow et al. (2025)). As $|m - m_\pm|$ grows and becomes comparable to or larger than $L_\pm$, Eq. (12) implies substantial decay in $D_p(m)$, so attenuation dominates and the log-odds response becomes strongly nonlinear and can collapse off-manifold.

**Fitting Form.** We fit the measured preference log-odds as a function of $m$ with

$$\log \frac{P(p_p \mid \tilde{h}(m))}{1 - P(p_p \mid \tilde{h}(m))} = (\alpha_p m + \beta_p)\, D_p(m) + b_p, \tag{15}$$

where $\beta_p = \boldsymbol{\omega}_p^\mathsf{T} \mathbf{h}$ is a per-example constant (since $\mathbf{h}$ is fixed for a given input), and $b_p$ is an offset.

**Fit Results.** Table 2 reports the fit quality of Eq. (15), with $R^2$ values exceeding 0.95 across most settings. These results validate the model's ability to accurately characterize the dynamics of preference log-odds. Details are in Appendix D.

### 4.3 Utility Capability: Only Validity Decay

**Utility Log-odds Under Manifold-Validity Decay.** Let $\mathbf{h} \in \mathbb{R}^{d_l}$ denote the activation at layer $l$. We quantify utility capability by the log-odds of positive vs. negative utility outcomes ($u_p/u_n$). Similar to preference, we assume utility is also associated with a direction $\boldsymbol{\omega}_u$ in activation space. Under steering $\tilde{h}(m) = h + m\,\Delta h$,, we model

$$\log \frac{P(u \mid \tilde{h}(m))}{1 - P(u \mid \tilde{h}(m))} = \boldsymbol{\omega}_u^{\mathsf{T}} \mathbf{h}\, D_u(m) + b_u, \quad (16)$$

where $D_u(m)$ follows the manifold-validity decay in Eq. (12) and decreases with $|m|$. Crucially, for *preference* steering directions we typically have $\boldsymbol{\omega}_u^{\mathsf{T}} \Delta \mathbf{h} \approx 0$, so utility is affected primarily through validity decay rather than a direct projection term.

**Fitting form.** Accordingly, we fit the measured utility log-odds with a pure decay curve:

$$\log \frac{P(u \mid \tilde{h}(m))}{1 - P(u \mid \tilde{h}(m))} = \beta_u\, D_u(m) + b_u, \quad (17)$$

where $\beta_u$ is the baseline log-odds and $b_u$ is an offset capturing residual bias.

**Fit Results.** Table 2 reports the fit quality of Eq. (17). Uniformly high $R^2$ values (typically $> 0.97$) suggest utility variations under preference steering are well captured by the proposed formulation. Additional details are in Appendix D.

| Type | Method | Preference $R^2 \uparrow$ | | | | Utility $R^2 \uparrow$ | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | PSY | PWR | AXB | **Avg** | PSY | PWR | AXB | **Avg** |
| ***Gemma-2-9B-IT*** | | | | | | | | | |
| Weight | SFT | 0.97 | 0.98 | 0.99 | 0.98 | 0.98 | 0.99 | 0.98 | 0.98 |
| | RePS | 0.99 | 0.99 | 0.99 | 0.99 | 0.96 | 0.99 | 0.99 | 0.98 |
| LoRA | SFT | 0.92 | 0.99 | 0.98 | 0.96 | 0.98 | 0.99 | 0.99 | 0.99 |
| | RePS | 0.83 | 0.99 | 0.99 | 0.94 | 0.99 | 0.99 | 0.99 | 0.99 |
| Vector | DiffMean | 0.97 | 0.99 | 0.99 | 0.98 | 0.97 | 0.99 | 0.98 | 0.98 |
| | SFT | 0.93 | 0.97 | 0.98 | 0.96 | 0.99 | 0.99 | 0.99 | 0.99 |
| | RePS | 0.99 | 0.98 | 0.95 | 0.97 | 0.99 | 0.99 | 0.99 | 0.99 |
| ***Qwen-2.5-7B-IT*** | | | | | | | | | |
| Weight | SFT | 0.99 | 0.99 | 0.99 | 0.99 | 0.99 | 0.99 | 0.98 | 0.99 |
| | RePS | 0.99 | 0.99 | 0.99 | 0.99 | 0.99 | 0.99 | 0.97 | 0.98 |
| LoRA | SFT | 0.97 | 0.99 | 0.99 | 0.98 | 0.99 | 0.99 | 0.99 | 0.99 |
| | RePS | 0.94 | 0.99 | 0.99 | 0.97 | 0.98 | 0.96 | 0.99 | 0.98 |
| Vector | DiffMean | 0.98 | 0.95 | 0.98 | 0.97 | 0.99 | 0.99 | 0.97 | 0.98 |
| | SFT | 0.93 | 0.97 | 0.98 | 0.96 | 0.98 | 0.99 | 0.99 | 0.99 |
| | RePS | 0.97 | 0.98 | 0.93 | 0.96 | 0.99 | 0.98 | 0.99 | 0.99 |

Table 2: **Curve fitting performance.** Results on Psychopathy (PSY), PowerSeeking (PWR), and AXBench (AXB). We report $R^2$ (higher is better), measuring alignment between theoretical curves and empirical data. Color intensity indicates $R^2$ values. Consistently dark shading shows high fidelity across settings ($R^2 > 0.95$).

## 5 Method

### 5.1 Preference–Utility Joint Optimization

Building on the preceding mechanistic analysis, we propose **S**teering with **P**reference–Uti**L**ity **I**nterven**T**ion (SPLIT), a training objective improving preference while delaying utility degradation.

**Utility Loss.** To preserve utility, we train on both the positive and negative samples for the same input using the language-modeling cross-entropy:

$$\mathcal{L}_{\text{util}} = \lambda_p\, \mathcal{L}_{\text{p}} + \lambda_n\, \mathcal{L}_{\text{n}}, \quad (18)$$

where $\mathcal{L}_{\text{p}}$ and $\mathcal{L}_{\text{n}}$ are the token-level cross-entropy losses on positive and negative samples, respectively, and $\lambda_p, \lambda_n$ control their relative weight.

**Preference Loss.** By Eq. (7), the loss gap $\mathcal{L}_{\text{n}} - \mathcal{L}_{\text{p}}$ is exactly the preference log-odds. We therefore maximize this gap via a hinge-style margin loss:

$$\mathcal{L}_{\text{pref}} = \gamma \cdot \sigma(\theta - (\mathcal{L}_{\text{n}} - \mathcal{L}_{\text{p}}))\,, \quad (19)$$

where $\sigma(\cdot)$ is ReLU and $\theta$ is a margin threshold, and $\gamma$ trades off preference improvement against utility preservation.

**Final Objective.** We combine the two components as

$$\mathcal{L} = \mathcal{L}_{\text{util}} + \mathcal{L}_{\text{pref}}. \quad (20)$$

### 5.2 Experiment Results.

We evaluate the proposed preference-utility joint optimization method under three intervention forms: local weight update, low-rank adaptation (LoRA), and activation vector steering. As shown in Table 3, our approach consistently achieves higher scores compared with baseline methods across all three intervention types. These results demonstrate the robustness and generality of the proposed optimization strategy.

## 6 Related Work

**Mechanism.** Most activation steering methods assume linear structure in activation space, controlling concepts by adding scaled direction vectors to hidden states (Mikolov et al., 2013; Pennington et al., 2014; Nanda et al., 2023; Tigges et al., 2023; Park et al., 2024; Wang et al., 2024; Yao et al.,

| Model | Form | Method | Psychopathy | PowerSeeking | AXBENCH | |
|---|---|---|---|---|---|---|
| | | | Acc(%, 0–100) ↑ | Concept(0–4) ↑ | Concept(0–2) ↑ | Harmonic(0–2) ↑ |
| Gemma-2-9B-IT | Vanilla | Vanilla | 50.00 | 1.87 | 0.4750 | 0.4950 |
| | Local Weight | SFT | **100.00** | 3.50 | 1.6625 | 1.4538 |
| | | REPS | **100.00** | 3.39 | 1.7750 | **1.6362** |
| | | SPLIT (Ours) | **100.00** | **3.59** | **1.8500** | 1.6225 |
| | LoRA | SFT | **100.00** | 3.41 | 1.7625 | 1.5188 |
| | | REPS | 99.00 | 3.44 | 1.7375 | **1.6525** |
| | | SPLIT (Ours) | **100.00** | **3.56** | **1.7750** | 1.6412 |
| | Vector | DiffMean | 53.00 | 2.95 | 1.1625 | 1.0550 |
| | | SFT | 97.00 | 3.30 | 1.7000 | 1.4487 |
| | | REPS | 98.00 | 3.61 | 1.7000 | 1.5550 |
| | | SPLIT (Ours) | **99.00** | **3.62** | **1.8500** | **1.6475** |
| Qwen-2.5-7B-IT | Vanilla | Vanilla | 50.00 | 2.24 | 0.4500 | 0.4713 |
| | Local Weight | SFT | 97.00 | 3.53 | 1.5375 | 1.1287 |
| | | REPS | 96.00 | 3.24 | 1.6875 | 1.4163 |
| | | SPLIT (Ours) | **98.00** | **3.66** | **1.7000** | **1.4325** |
| | LoRA | SFT | 99.00 | 3.05 | 1.4875 | 1.3175 |
| | | REPS | **100.00** | 3.34 | 1.4875 | 1.4013 |
| | | SPLIT (Ours) | **100.00** | **3.59** | **1.7375** | **1.6362** |
| | Vector | DiffMean | 55.00 | 3.17 | 0.9500 | 0.9950 |
| | | SFT | 97.00 | 3.58 | 1.5750 | 1.5800 |
| | | REPS | 88.00 | 3.63 | 1.7375 | 1.6412 |
| | | SPLIT (Ours) | **98.00** | **3.65** | **1.8125** | **1.6500** |

Table 3: Main task performance of steering methods evaluated on three datasets. *Psychopathy* is reported with classification accuracy (Acc, %), *PowerSeeking* is evaluated using LLM-judge preference scores on a 0–4 scale, and *AxBench* reports the concept score and the harmonic mean (HM) over concept, instruction, and fluency scores, each on a 0–2 scale, as evaluated by an LLM judge. All methods perform inference-time interventions on hidden representations. **Best** and second-best results are highlighted within each model and intervention form.

2025; Zhang et al., 2026; Hu et al., 2026). Building on this view, Bigelow et al. (2025) show that steering yields an approximately linear trend in posterior odds, but mainly in the small-scale regime. Recent studies further report non-monotonic or adverse effects under stronger steering, challenging a naive global linearity assumption (Bricken et al., 2023; Wollschläger et al., 2025). Meanwhile, representation-manifold work provides a complementary geometric lens for understanding steering and its limitations (Modell et al., 2025; Li and He, 2025; Xie et al., 2025).

**Activation Steering.** Activation steering controls the behavior of LLMs by intervening in hidden states during forward propagation, using steering vectors to control single attributes as well as more complex behavioral targets (Turner et al., 2023; Rimsky et al., 2024; van der Weij et al., 2024; Rahn et al., 2024; Scalena et al., 2024; Tan et al., 2024; Bhattacharjee et al., 2024; Postmus and Abreu, 2024; Konen et al., 2024; Hazra et al., 2024; Han et al., 2025; Jiang et al., 2025). However, recent studies have shown that the coarse-grained nature of activation steering can lead to degradation in model utility (Wang et al., 2025; Wu et al., 2025a). Cao et al. (2024); Wu et al. (2025b) introduce explicit preference learning objectives to optimize activation steering, enabling more precise control.

**Parameter-Efficient Fine-Tuning.** Parameter-Efficient Fine-Tuning (PEFT) methods, including adapters and LoRA, show that effective adaptation of LLMs does not require updating all parameters. LoRA achieves performance comparable to full fine-tuning, indicating that adaptation relies on structured low-rank weight updates rather than full parameter changes (Hu et al., 2022a; Zi et al., 2023; Zhang et al., 2023; Hayou et al., 2024; Kopiczko et al., 2024; Zhang et al., 2024). Local weight updates further reveal that LLM knowledge is highly localized, as modifying a small subset of parameters in specific layers suffices to change factual associations (Zaken et al., 2022; Ding et al., 2022; Geva et al., 2021; Yang et al., 2025).

## 7 Conclusion

We propose a unified dynamic weight update framework that incorporates parameter updates, LoRA, and activation interventions, revealing a consistent preference–utility decay pattern in the log-odds space. Building on this mechanistic insight, we de-